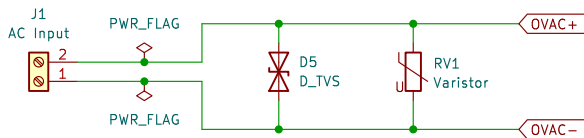
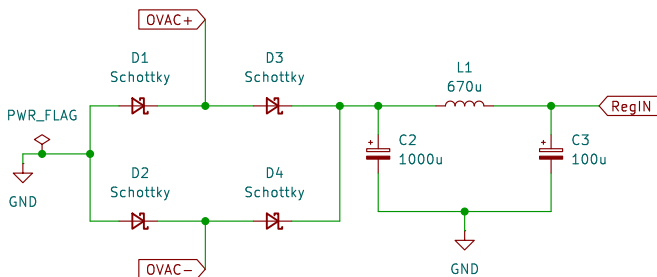


AC to DC Regulated Power Supply with Overvoltage Protection

Input Power stage & Overvoltage Protection



Bridge Rectification & CLC Filter



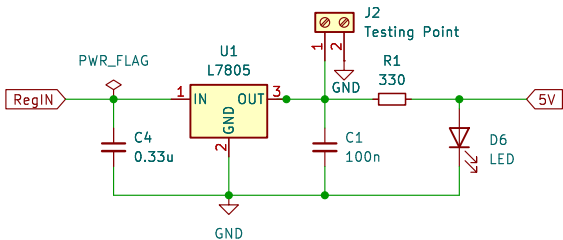
Electrical Specifications

Parameter	Value
Input Voltage	9-12VAC
Output Voltage	5.0 VDC ($\pm 4\%$)
Max Output Current	1.0 A
Ripple Rejection	CLC Pi-Filter + Linear Regulation

Bill of Materials

Component Category	Specification	Qty
Rectification	SS34 Schottky Barrier Diodes (3A)	4
Filtration (Pi-Filter)	CD127 Inductor	1
	Low ESR Caps	3
Regulation	LM7805 LDO	1
Connectors & Misc	Screw Terminals, LED, Resistors	-
Input Protection	MOV (14D)	1
	PPTC Fuse (0.5A)	1
	TVS Diode	1

Regulated DC



SB Karthik

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Title: Regulated Power Supply

Size: A4
KiCad E.D.A. 9.0.0

Date: 2026-01-01

Rev: 2.2

Id: 1/1